a THE PROBLEM

Robustness fine-tuning teaches VLMs to answer the unanswerable



VizWiz photo

Q: *"What flavor of jellybeans are in this package?"*

Label: **unanswerable**

Corrupted-image accuracy

RKL +2.0 **GRPO +8.5**

but unanswerable accuracy

76.0 → 56.5 / 60.3

Base (Qwen3-VL-4B) ✓

"The image is too blurry and the text is not legible …" → **unanswerable**

+ RKL distillation *reasoning hallucinates* ✗

"The flavors visible are Cherry, Lime, Orange, and Strawberry … clearly visible." → **Cherry, Lime, Orange, Strawberry**

+ GRPO *answer ignores reasoning* ✗

"… the labels are illegible. Therefore, the answer is unanswerable. " → **strawberry**

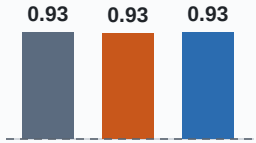
Verbatim response excerpts, greedy decoding.

b DIAGNOSIS

The model still knows; training breaks a different step

1 Linear probe on hidden states

Does the model encode "unanswerable"? (AUROC)



Unchanged after fine-tuning; the probe flags **77%** of the hallucinated answers.

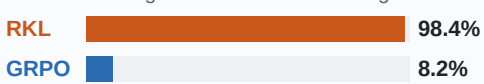
Knows, but answers anyway.

2 Answer-commitment test

Give each model the base model's "can't tell" reasoning: does its answer follow?



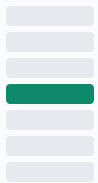
Refuses when given "can't tell" reasoning



RKL: the reasoning goes wrong. **GRPO:** the answer step ignores the reasoning.

c OUR FIX

Training-free contrastive steering restores abstention

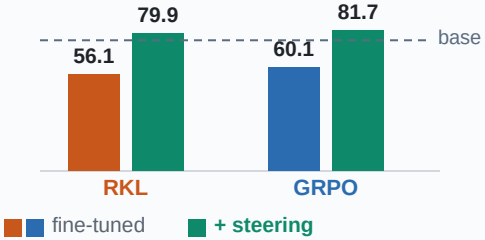


v = h(**refusal**) − h(**answer**)
averaged over refusal-vs-answer pairs

$h\ell \leftarrow h\ell + \lambda \cdot v$

optional gate: entropy > τ

Unanswerable VizWiz accuracy (%)



GRPO answerable **63.8** → **68.7** ; corrupted accuracy within ±0.5

GRPO + steering ✓

"The image is too blurry to determine the flavor …" → **unanswerable**